

Lab: Sensory Neuroscience -- How Neurons Turn the World into Predictions

Objective

Investigate how the nervous system processes sensory signals through **prediction and error correction**, examining the pathway from receptor cells to cortical processing.

Prerequisites

- Completed Biology modules on Systems and Agents
- Read the module on How Neurons Process Sensory Signals
- Basic knowledge of neuron structure (axons, dendrites, synapses)

Part 1: Sensory Pathway Mapping

Goal: Trace how a sensory signal becomes a perception through biological prediction.

Choose one sensory system (vision, hearing, touch, taste, or smell). Research and map the pathway:

Stage	Structure	What Happens	Active Inference Role
1. Stimulus reception			Sensory state
2. Transduction (signal conversion)			Encoding observations
3. Relay station			
4. Primary cortex processing			Prediction comparison
5. Higher cortex interpretation			Generative model output

Write your response here

Part 2: The Predictive Brain Experiment

Goal: Experience how your nervous system uses top-down predictions.

Experiment A -- Sensory adaptation: Stare at a colored square for 30 seconds, then look at a white surface. Record the afterimage color. - Why does your brain produce an afterimage? (Hint: neurons adapted to the color reduce their firing rate, and the brain's prediction is now "wrong.")

Experiment B -- Two-point discrimination: Using two pencil tips held close together, touch your fingertip, then your forearm. Can you feel two points in both places? - Why is your fingertip better at telling them apart? (Hint: receptor density = precision of sensory signals.)

Record your results:

Experiment	What Happened	Biological Explanation	Active Inference Connection
A: Afterimage			
B: Two-point			

Write your response here

Part 3: Predictive Coding in the Visual System

Goal: Understand how the visual cortex implements predictive coding.

Research the hierarchy of visual processing (V1, V2, V4, IT cortex). Answer:

1. What does the primary visual cortex (V1) detect? (Edges, orientations.)
2. What do higher areas detect? (Shapes, objects, faces.)
3. In predictive coding, higher areas send **predictions** down to lower areas. Lower areas send **prediction errors** up. Draw or describe this flow.
4. How does this explain why you can recognize a friend's face in a crowd almost instantly?

Write your response here

Part 4: Sensory Disorders as Prediction Failures

Goal: Analyze a sensory condition through the Active Inference lens.

Choose one: phantom limb pain, tinnitus, synesthesia, or color blindness.

- 1. What is the condition?
- 2. What is the sensory prediction error? (What does the brain expect vs. what does it receive?)
- 3. Why can the brain not resolve this error through normal perception?
- 4. How might treatments work by correcting the brain's predictions?

Write your response here

Summary Table

Concept	Biological Mechanism	Active Inference Term
Sensory transduction	Receptor cells convert stimuli to electrical signals	Observation encoding
Predictive coding	Cortical hierarchy sends predictions down and errors up	Hierarchical inference
Sensory precision	Receptor density and signal-to-noise ratio	Precision weighting
Sensory disorders	Mismatch between predictions and available sensory input	Persistent prediction error